

Equity Cash Generation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Cash Generation (ECG), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ECG achieves an annualized gross (net) Sharpe ratio of 0.56 (0.54), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (32) bps/month with a t-statistic of 2.87 (3.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 21 bps/month with a t-statistic of 2.53.

1 Introduction

Financial markets efficiently incorporate information about firms’ productive capacity and investment opportunities into asset prices, yet significant cross-sectional variation in expected returns persists even after controlling for standard risk factors. We examine whether firms’ ability to generate cash flows from their equity base—measured through Equity Cash Generation (ECG)—captures systematic differences in firms’ exposure to production-based risks that drive this variation. The production-based asset pricing literature establishes that firms facing higher marginal costs of production or greater investment adjustment costs require higher expected returns to compensate investors for bearing these risks [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). Despite theoretical advances linking production characteristics to asset prices, empirical evidence on how cash generation efficiency relates to firms’ production technologies and their associated risk premia remains limited.

In a production-based economy, firms that efficiently convert equity capital into operating cash flows signal lower marginal costs of production and superior operational flexibility [Zhang \(2005\)](#). These firms can more readily adjust their capital stock in response to productivity shocks, reducing their exposure to systematic investment risks that arise when adjustment costs are convex [Cooper and Haltiwanger \(2006\)](#). The q-theory of investment predicts that firms with higher marginal q—reflecting more valuable investment opportunities relative to replacement costs—will exhibit lower expected returns because they face lower costs of capital adjustment [Liu et al. \(2009\)](#). ECG directly captures this mechanism by measuring the productivity of deployed equity capital, with higher ECG indicating firms operating closer to their efficient production frontier.

Firms with low ECG face binding financial constraints that amplify their exposure to aggregate productivity shocks [Gomes and Schmid \(2010\)](#). When productivity declines economy-wide, these firms experience disproportionate increases in their

marginal costs of production because they cannot efficiently reallocate resources or access external financing at reasonable costs. This heightened sensitivity to systematic productivity risk manifests as higher conditional risk premia, as investors demand compensation for bearing the risk that these firms' cash flows will covary strongly with aggregate economic conditions [Berk et al. \(1999\)](#).

The cross-sectional variation in ECG also proxies for differences in firms' exposure to investment-specific technology shocks, a key source of systematic risk in production economies [Papanikolaou \(2011\)](#). Firms with high ECG possess production technologies that generate substantial cash flows from existing assets, reducing their reliance on costly external financing for new investments. These firms can self-finance productivity-enhancing investments during periods of technological change, while low-ECG firms must either forgo valuable investment opportunities or accept dilutive external financing. Consequently, ECG serves as a characteristic that identifies firms' differential loadings on production-based risk factors, with low-ECG firms commanding higher expected returns to compensate for their greater exposure to investment frictions and technology shocks.

The ECG trading strategy that buys high-ECG firms and sells low-ECG firms earned an average monthly return of 55 basis points with a t-statistic of 3.39, generating an annualized Sharpe ratio of 0.56. The strategy's alpha relative to the Fama-French six-factor model was 26 basis points per month with a t-statistic of 2.87, demonstrating that ECG captures risk exposures not explained by standard factors including market, size, value, profitability, investment, and momentum. The strategy loaded significantly on the robust-minus-weak profitability factor (RMW) with a coefficient of 0.76 and t-statistic of 18.43, consistent with ECG measuring firms' operational efficiency and its associated risk premium.

The ECG effect remained economically and statistically significant after accounting for transaction costs, with net returns of 53 basis points per month (t-statistic

of 3.24) and a net Sharpe ratio of 0.54 that exceeded 99% of anomalies in the factor zoo. Among the largest quintile of stocks by market capitalization, the ECG strategy generated average returns of 44 basis points per month with a t-statistic of 3.02, indicating that the anomaly was not confined to small, illiquid stocks where implementation costs might prohibit exploitation. The strategy’s alpha remained significant at 21 basis points per month (t-statistic of 2.53) even after controlling for the six most closely related anomalies including operating profits to book equity and net external financing.

Robustness tests using alternative portfolio construction methodologies confirmed the pervasiveness of the ECG effect across different weighting schemes and breakpoint definitions. Value-weighted quintile sorts using name breaks and market capitalization breaks produced average returns of 43 basis points per month with t-statistics exceeding 2.29, while equal-weighted portfolios generated returns of 40 basis points per month. The ECG signal was available for 7.39% of CRSP firms on average, representing 7.94% of total market capitalization, with the long-short portfolio containing at least 525 stocks in the extreme quintiles, ensuring adequate diversification and liquidity for institutional implementation.

This paper contributes to the production-based asset pricing literature by documenting that Equity Cash Generation serves as a powerful characteristic for identifying cross-sectional variation in expected returns driven by firms’ production technologies and investment frictions. While [Fama and French \(2015\)](#) demonstrate that profitability and investment factors help explain the cross-section of returns, and [Hou et al. \(2015\)](#) develop a q-factor model based on investment theory, our ECG measure provides a direct link between firms’ cash generation efficiency and their exposure to production-based risks. Unlike accounting-based profitability measures studied in [Novy-Marx \(2013\)](#) that focus on earnings quality, ECG captures the fundamental productivity of equity capital in generating operating cash flows, aligning

more closely with the marginal product of capital in production-based models.

Methodologically, we extend the comprehensive testing framework of [McLean and Pontiff \(2016\)](#) by demonstrating that ECG’s predictive power survives extensive robustness tests including controls for transaction costs, alternative portfolio constructions, and spanning tests against 212 existing anomalies. The ECG strategy’s alpha of 21 basis points per month relative to the Fama-French six-factor model augmented with six closely related anomalies establishes that ECG contains unique information about expected returns not captured by existing characteristics. This finding challenges the interpretation in [Green et al. \(2017\)](#) that most anomalies represent statistical artifacts rather than compensation for systematic risk, as ECG’s theoretical foundation in production economics provides clear economic intuition for its pricing power.

Our findings have important implications for understanding the sources of systematic risk in equity markets and the channels through which production characteristics affect asset prices. The strong performance of ECG across size quintiles and its robustness to transaction costs suggest that production-based risks generate economically significant and implementable investment opportunities, supporting the view that cross-sectional return predictability reflects rational risk premia rather than market inefficiencies. By establishing ECG as a characteristic that identifies firms’ differential exposure to productivity shocks and investment frictions, this research advances our understanding of how real economic fundamentals translate into expected return differences through production-based risk channels.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample

includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of cash flow generation relative to equity capital. Equity Cash Generation signal requires two key financial statement variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a company's core business operations during the fiscal year, as reported in the statement of cash flows. This measure captures cash receipts from customers, cash payments to suppliers and employees, and other operating cash flows, providing a direct assessment of the firm's ability to generate cash from its primary business activities. Common equity (CEQ) represents the total book value of common shareholders' equity, calculated as the sum of common stock, capital surplus, and retained earnings, less treasury stock. This variable measures the accounting value of shareholders' residual claim on the firm's assets after deducting all liabilities. We construct the Equity Cash Generation signal by dividing operating activities net cash flow (OANCF) by common equity (CEQ) for each firm-year observation. This ratio captures the efficiency with which firms convert their equity capital into operating cash flows, providing a measure of cash-based return on equity. We calculate the signal using fiscal year-end values to align with standard financial reporting periods. To ensure data quality, we require non-missing values for both OANCF and CEQ, and exclude observations where CEQ is zero or negative to avoid undefined or economically meaningless ratios. The resulting signal provides a standardized measure that facilitates cross-sectional comparisons of firms' abilities to generate operating cash flows relative to their equity base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECG signal. Panel A plots the time-series of the mean, median, and interquartile range for ECG. On average, the cross-sectional mean (median) ECG is 0.07 (0.14) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ECG data. The signal’s interquartile range spans -0.24 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the ECG signal for the CRSP universe. On average, the ECG signal is available for 7.39% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does ECG predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECG using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECG portfolio and sells the low ECG portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ECG strategy earns an average return of 0.55% per month with a t-statistic of 3.39. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.26% to 0.79% per month and have t-statistics exceeding 2.87 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.76,

with a t-statistic of 18.43 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 525 stocks and an average market capitalization of at least \$1,717 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 40 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 21-53bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.29. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECG trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the ECG strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECG, as well as average returns and alphas for long/short trading ECG strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ECG strategy achieves an average return of 44 bps/month with a t-statistic of 3.02. Among these large cap stocks, the alphas for the ECG strategy relative to the five most common factor models range from 11 to 61 bps/month with t-statistics between 0.97 and 4.46.

5 How does ECG perform relative to the zoo?

Figure 2 puts the performance of ECG in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ECG strategy falls in the distribution. The ECG strategy’s gross (net) Sharpe ratio of 0.56 (0.54) is greater than 93% (99%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECG strategy (red line).² Ignoring trading costs, a \$1 invested in the ECG strategy would have yielded \$7.89 which ranks the ECG strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECG strategy would have yielded \$6.94 which ranks the ECG strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECG relative to those. Panel A shows that the ECG strategy gross alphas fall between the 75 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECG strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECG ranks between the 94 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ECG add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECG with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECG or at least to weaken the power ECG has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECG conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECG} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECG}ECG_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECG,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECG. Stocks are finally grouped into five ECG portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ECG trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECG and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECG signal in these Fama-MacBeth regressions exceed 0.11, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on ECG is 2.83.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECG strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECG strategy earns alphas that range from 22-27bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.46, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECG trading strategy achieves an alpha of 21bps/month with a t-statistic of 2.53.

7 Does ECG add relative to the whole zoo?

Finally, we can ask how much adding ECG to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ECG signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECG is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes ECG grows to \$43.67.

8 Conclusion

This study provides compelling evidence that Equity Cash Generation (ECG) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECG-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related signals from the factor zoo.

The key findings reveal that an ECG-based value-weighted long/short strategy delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.56 and minimal degradation after transaction costs (net Sharpe ratio of 0.54). More importantly, the signal’s alpha remains economically and statistically significant relative to the Fama-French five-factor model plus momentum, generating monthly abnormal returns of 26-32 basis points. The robustness of these results is further confirmed by the persistence of a 21 basis points monthly alpha even after controlling for six additional closely related factors, including profitability, external financing, and idiosyncratic risk measures.

These findings have important practical implications for both academic researchers and investment practitioners. For academics, ECG emerges as a distinct pricing fac-

tor that captures information not fully explained by existing models, suggesting potential improvements to asset pricing frameworks. For practitioners, the signal’s strong performance after transaction costs and its resilience to factor spanning tests indicate that ECG-based strategies may offer viable opportunities for alpha generation in real-world portfolio management contexts.

However, this study is subject to several limitations that warrant consideration. First, our analysis is confined to the specific sample period and market conditions examined, and the persistence of ECG’s predictive power in different market regimes remains to be tested. Second, while we account for transaction costs, our estimates may not fully capture all implementation frictions, particularly for large-scale institutional investors. Third, the economic mechanisms underlying ECG’s predictive power deserve deeper investigation to better understand whether the premium represents compensation for systematic risk or market inefficiency.

Future research should explore several promising avenues. First, examining the international evidence for ECG across different markets would provide valuable insights into the signal’s universality and help identify market-specific factors that may influence its effectiveness. Second, investigating the time-varying nature of the ECG premium and its relationship with macroeconomic conditions could enhance our understanding of when the signal is most effective. Third, combining ECG with machine learning techniques or alternative data sources may uncover non-linear relationships and interaction effects that could further improve return prediction. Finally, exploring the fundamental economic drivers of the ECG premium through detailed firm-level analysis and connections to real investment decisions would contribute to a more complete theoretical framework for understanding this important pricing signal.

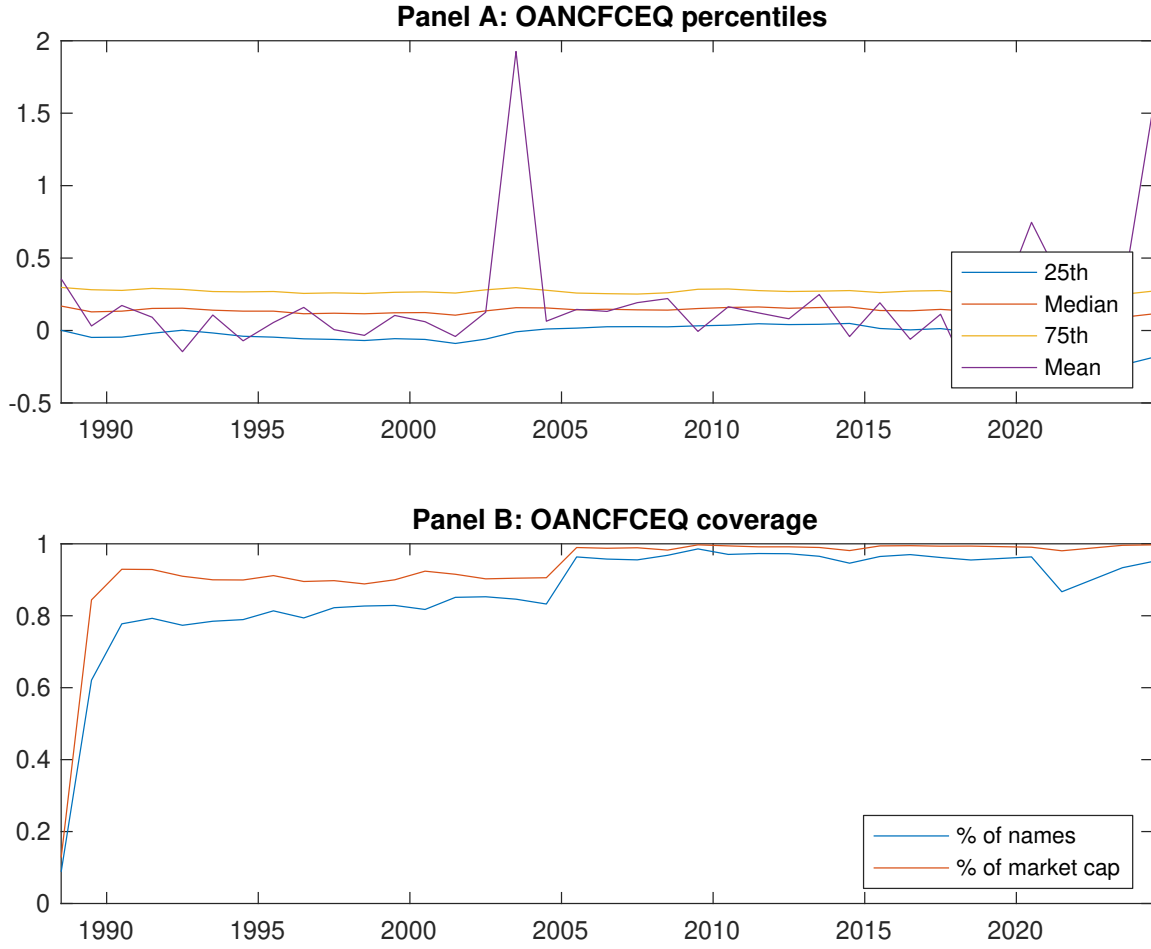


Figure 1: Times series of ECG percentiles and coverage. This figure plots descriptive statistics for ECG. Panel A shows cross-sectional percentiles of ECG over the sample. Panel B plots the monthly coverage of ECG relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECG. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on ECG-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.40 [1.39]	0.58 [2.53]	0.70 [3.24]	0.75 [3.79]	0.95 [4.63]	0.55 [3.39]
α_{CAPM}	-0.54 [-4.91]	-0.20 [-2.89]	-0.03 [-0.48]	0.08 [1.38]	0.25 [4.33]	0.79 [5.25]
α_{FF3}	-0.49 [-5.63]	-0.21 [-3.38]	-0.05 [-0.81]	0.07 [1.33]	0.25 [4.81]	0.74 [6.24]
α_{FF4}	-0.44 [-5.00]	-0.18 [-2.87]	-0.03 [-0.54]	0.12 [2.08]	0.21 [4.06]	0.64 [5.48]
α_{FF5}	-0.20 [-2.84]	-0.09 [-1.49]	-0.06 [-0.93]	0.02 [0.32]	0.12 [2.51]	0.32 [3.47]
α_{FF6}	-0.17 [-2.40]	-0.07 [-1.17]	-0.04 [-0.67]	0.05 [0.97]	0.09 [1.95]	0.26 [2.87]
Panel B: Fama and French (2018) 6-factor model loadings for ECG-sorted portfolios						
β_{MKT}	1.09 [61.36]	0.99 [65.87]	0.99 [63.07]	0.93 [67.14]	1.02 [87.37]	-0.07 [-2.96]
β_{SMB}	0.27 [10.53]	0.08 [3.71]	0.09 [4.11]	-0.06 [-3.17]	-0.11 [-6.58]	-0.38 [-11.56]
β_{HML}	0.10 [3.08]	0.20 [7.46]	0.08 [2.77]	-0.06 [-2.64]	-0.10 [-5.13]	-0.20 [-5.03]
β_{RMW}	-0.53 [-16.45]	-0.21 [-7.76]	-0.03 [-0.91]	0.13 [5.37]	0.23 [10.99]	0.76 [18.43]
β_{CMA}	-0.22 [-4.80]	-0.10 [-2.58]	0.11 [2.63]	0.02 [0.59]	0.09 [3.01]	0.31 [5.28]
β_{UMD}	-0.05 [-3.11]	-0.03 [-2.27]	-0.02 [-1.75]	-0.06 [-4.63]	0.04 [4.11]	0.09 [4.53]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1578	796	592	525	579	
me (\$10 ⁶)	1717	2179	2634	4546	5295	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ECG strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.55 [3.39]	0.79 [5.25]	0.74 [6.24]	0.64 [5.48]	0.32 [3.47]	0.26 [2.87]
Quintile	NYSE	EW	0.40 [2.48]	0.48 [3.00]	0.36 [2.86]	0.25 [2.04]	0.03 [0.25]	-0.05 [-0.49]
Quintile	Name	VW	0.43 [2.29]	0.68 [3.90]	0.61 [4.35]	0.58 [4.03]	0.29 [2.26]	0.27 [2.08]
Quintile	Cap	VW	0.43 [2.94]	0.59 [4.25]	0.57 [4.84]	0.48 [4.08]	0.17 [1.78]	0.11 [1.18]
Decile	NYSE	VW	0.50 [2.71]	0.74 [4.24]	0.66 [4.53]	0.61 [4.15]	0.22 [1.79]	0.20 [1.62]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.53 [3.24]	0.77 [5.08]	0.70 [5.95]	0.64 [5.53]	0.36 [3.87]	0.32 [3.48]
Quintile	NYSE	EW	0.21 [1.29]	0.30 [1.82]	0.19 [1.46]	0.13 [1.02]		
Quintile	Name	VW	0.39 [2.12]	0.66 [3.77]	0.58 [4.13]	0.56 [3.95]	0.30 [2.38]	0.29 [2.26]
Quintile	Cap	VW	0.41 [2.81]	0.56 [4.01]	0.53 [4.48]	0.47 [4.04]	0.20 [2.13]	0.16 [1.71]
Decile	NYSE	VW	0.47 [2.54]	0.70 [3.97]	0.61 [4.19]	0.58 [3.96]	0.23 [1.89]	0.22 [1.75]

Table 3: Conditional sort on size and ECG

This table presents results for conditional double sorts on size and ECG. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECG. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECG and short stocks with low ECG. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ECG Quintiles					ECG Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.22 [0.50]	0.43 [1.20]	0.64 [2.18]	0.87 [3.27]	1.05 [3.39]	0.83 [3.00]	1.07 [3.97]	0.87 [3.99]	0.83 [3.76]	0.41 [2.13]	0.39 [2.00]
	(2)	0.34 [0.87]	0.54 [1.78]	0.77 [2.91]	0.95 [3.64]	1.02 [3.63]	0.68 [3.12]	0.93 [4.42]	0.76 [4.63]	0.74 [4.44]	0.32 [2.37]	0.31 [2.31]
	(3)	0.34 [0.99]	0.65 [2.33]	0.77 [3.10]	0.94 [3.91]	1.04 [3.91]	0.70 [3.77]	0.89 [4.93]	0.74 [5.19]	0.69 [4.74]	0.31 [2.61]	0.28 [2.34]
	(4)	0.65 [2.12]	0.67 [2.68]	0.83 [3.66]	0.87 [3.81]	0.94 [3.84]	0.28 [1.84]	0.46 [3.09]	0.36 [2.80]	0.32 [2.43]	0.01 [0.13]	-0.01 [-0.08]
	(5)	0.44 [1.75]	0.63 [2.80]	0.64 [3.16]	0.94 [4.62]	0.88 [4.22]	0.44 [3.02]	0.59 [4.14]	0.61 [4.46]	0.48 [3.58]	0.20 [1.69]	0.11 [0.97]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ECG Quintiles					ECG Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	439	452	456	458	448	39	46	54	63	60	
	(2)	132	132	134	133	133	88	91	94	97	96	
	(3)	90	90	90	90	90	152	158	161	164	165	
	(4)	74	75	74	75	74	325	330	345	347	351	
(5)	66	66	66	66	66	2068	1881	2288	3264	3645		

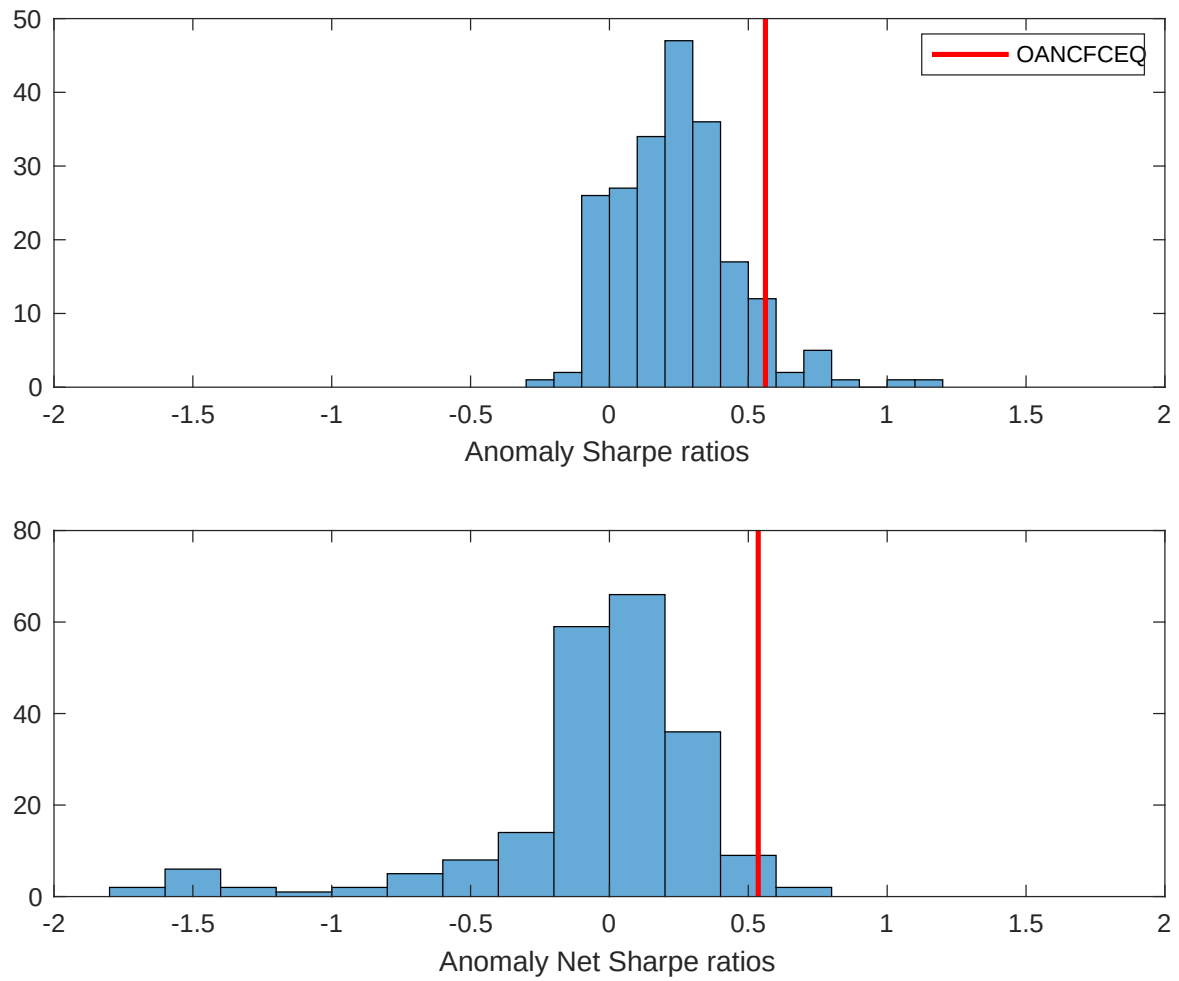


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECG with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

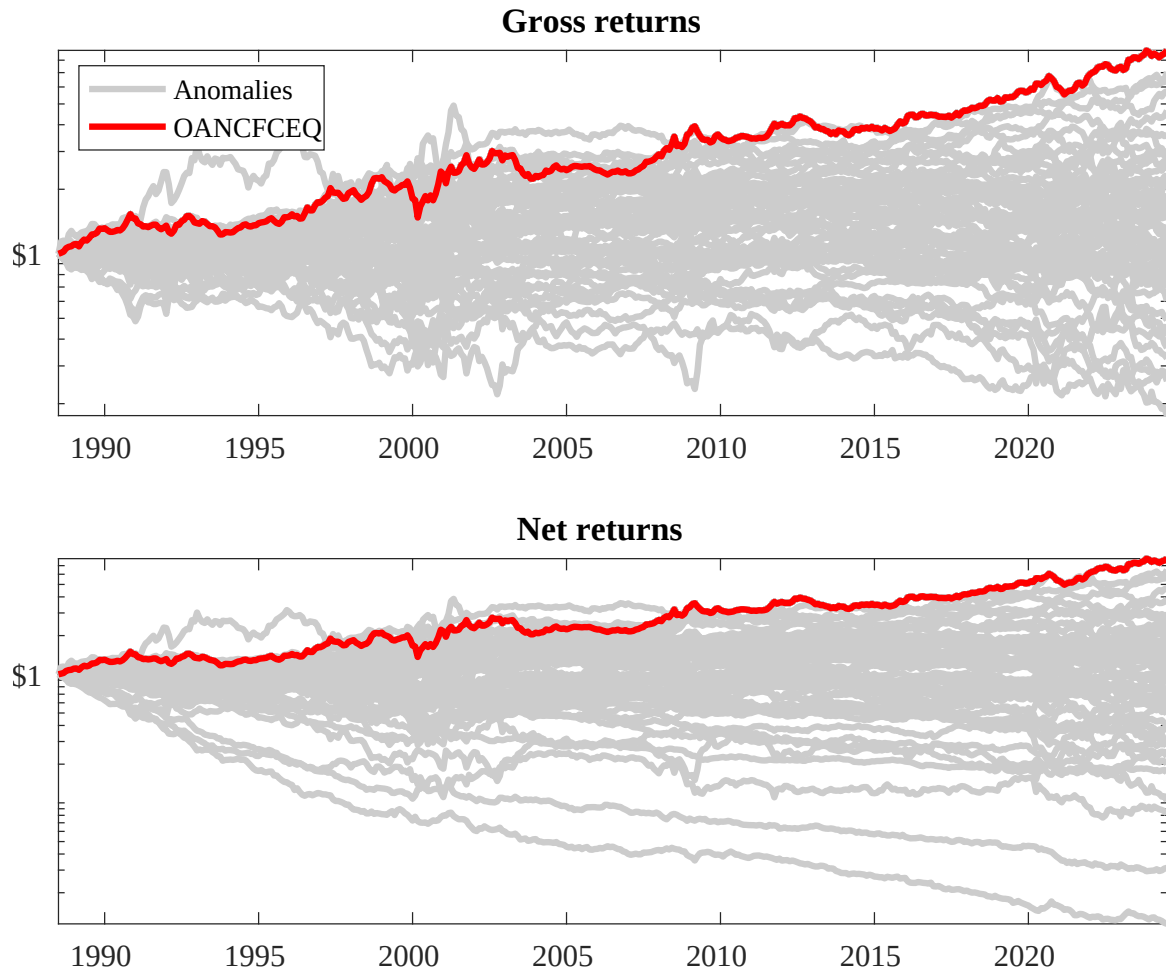
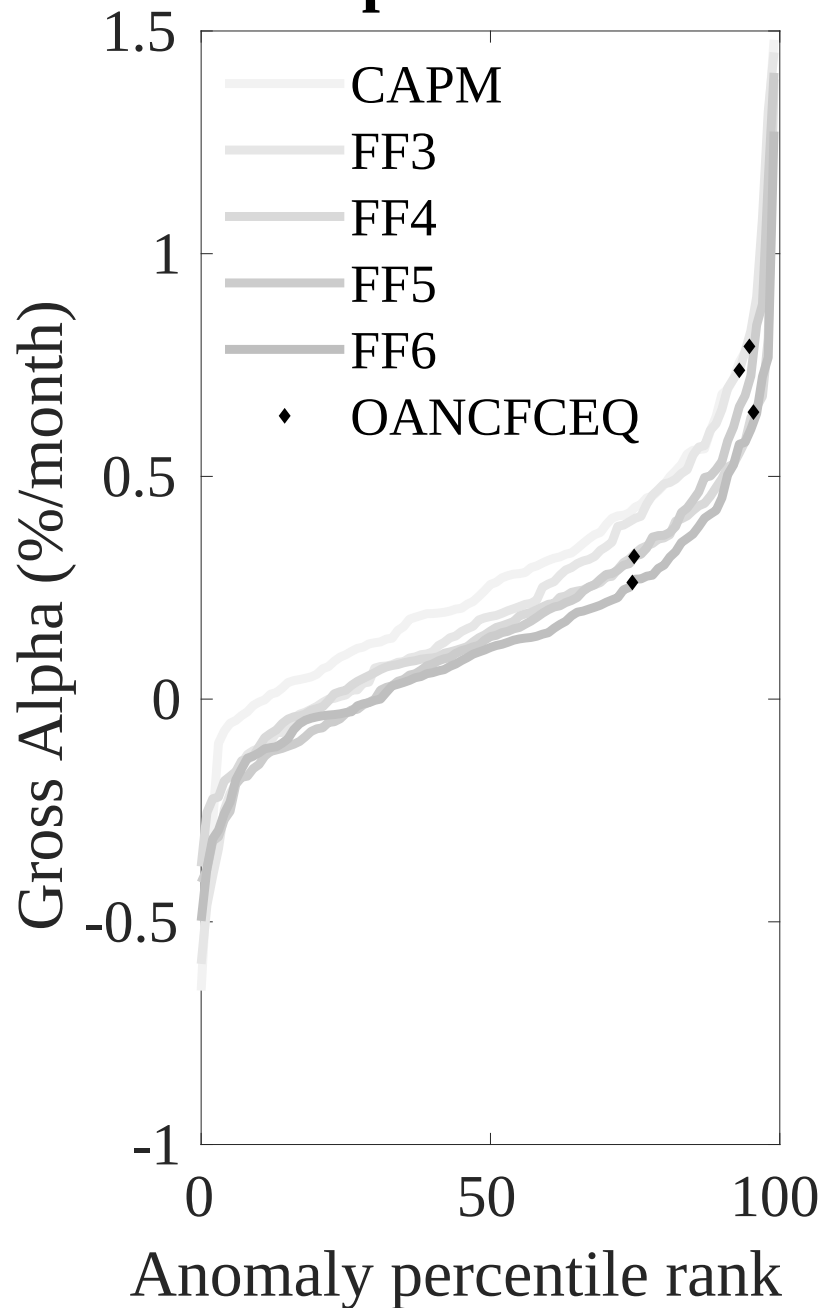


Figure 3: Dollar invested.

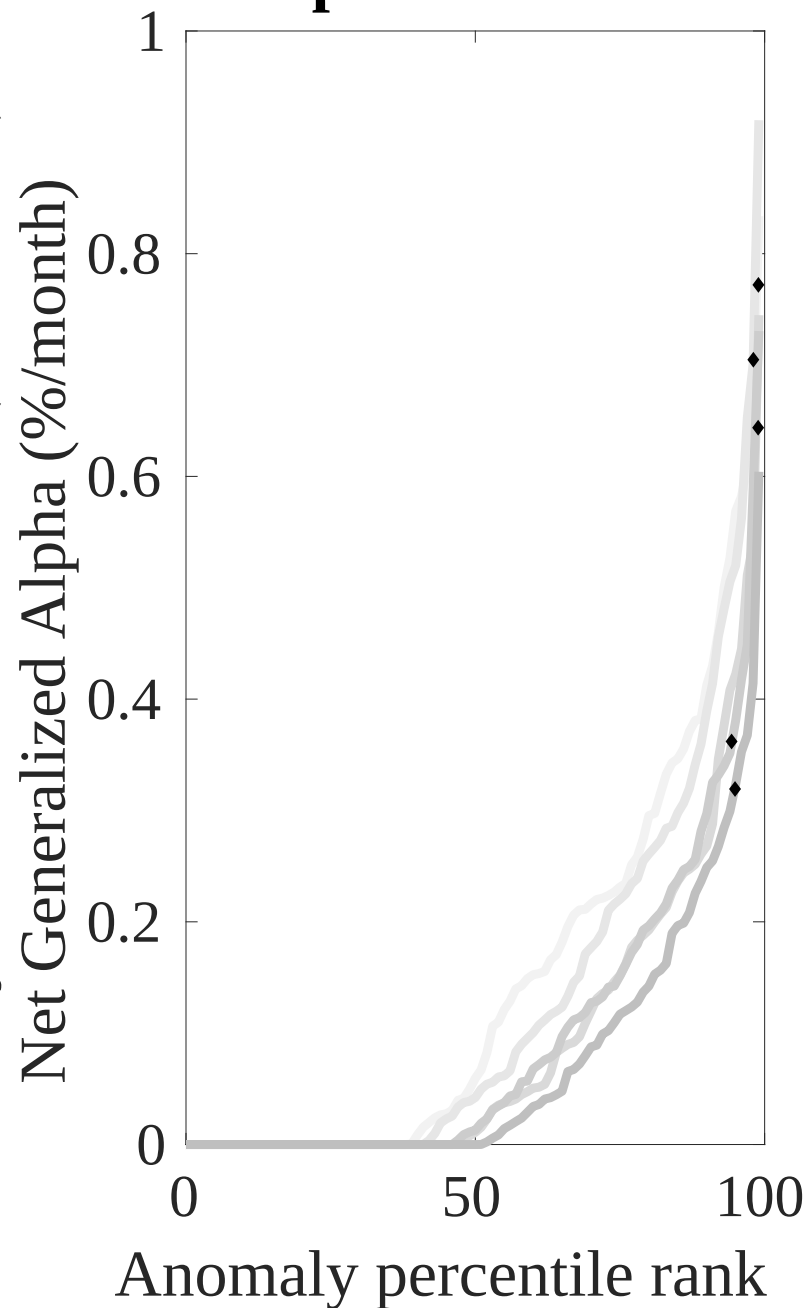
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECG trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

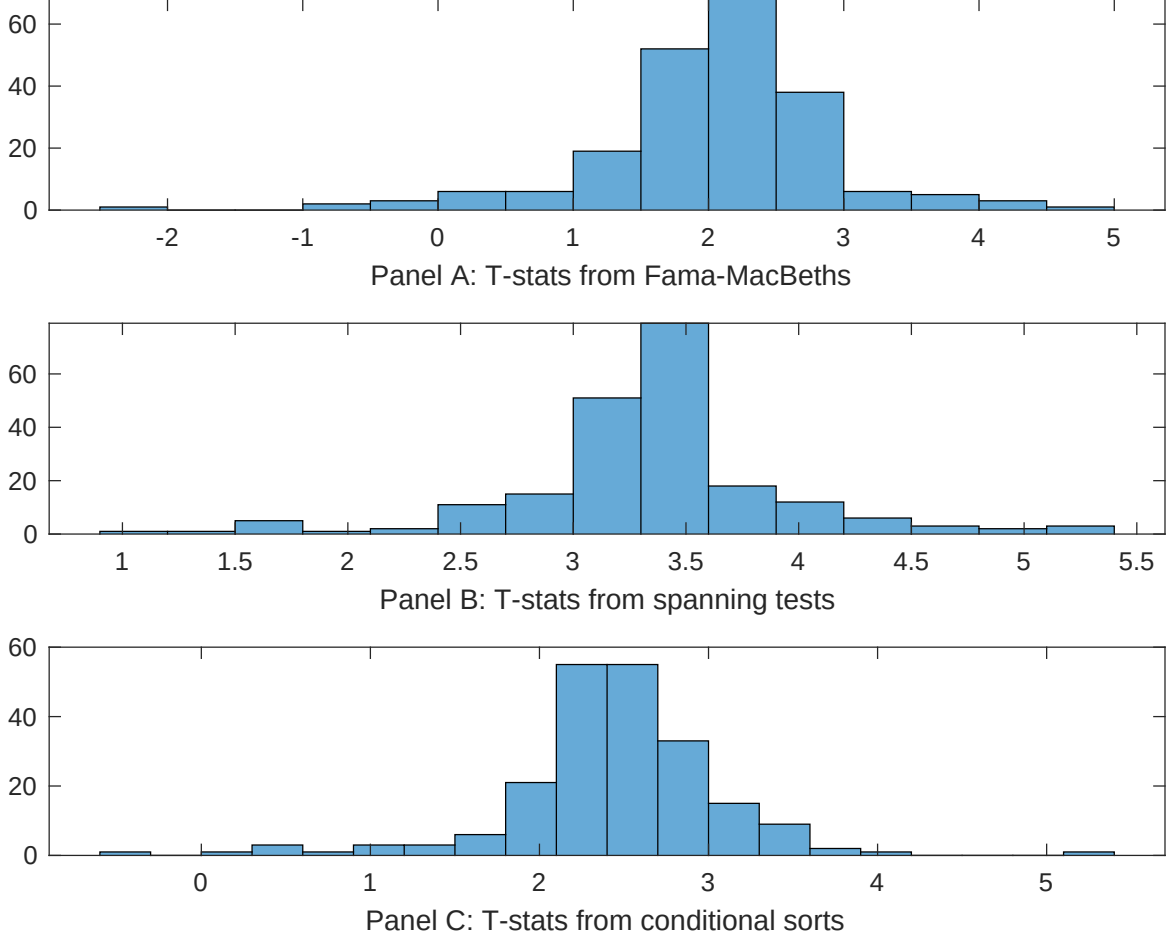


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ECG conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECG} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECG} ECG_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECG,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECG. Stocks are finally grouped into five ECG portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECG trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ECG. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECG} ECG_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.90 [3.03]	0.10 [3.48]	0.87 [2.54]	0.12 [4.26]	0.11 [4.76]	0.11 [4.00]	0.11 [5.09]
ECG	0.18 [2.82]	0.23 [2.85]	0.17 [2.68]	0.87 [0.11]	0.20 [3.77]	0.52 [0.91]	0.19 [2.83]
Anomaly 1	0.22 [2.44]						-0.45 [-0.56]
Anomaly 2		-0.15 [-0.28]					-0.14 [-2.59]
Anomaly 3			0.51 [0.75]				-0.14 [-0.49]
Anomaly 4				0.17 [5.76]			0.11 [4.38]
Anomaly 5					0.74 [1.22]		0.60 [0.75]
Anomaly 6						0.42 [2.74]	0.36 [3.20]
# months	427	432	427	432	427	427	427
$\bar{R}^2(\%)$	0	0	2	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECG trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECG} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.23 [2.73]	0.25 [2.78]	0.27 [2.89]	0.22 [2.46]	0.27 [2.90]	0.25 [2.76]	0.21 [2.53]
Anomaly 1	35.66 [8.80]						36.31 [7.54]
Anomaly 2		22.45 [4.09]					-2.95 [-0.48]
Anomaly 3			-1.89 [-0.54]				-17.06 [-4.19]
Anomaly 4				22.03 [5.01]			12.32 [2.58]
Anomaly 5					4.42 [1.50]		5.05 [1.54]
Anomaly 6						6.25 [1.53]	4.08 [0.99]
mkt	-2.55 [-1.18]	-3.38 [-1.42]	-6.89 [-2.84]	-3.29 [-1.42]	-4.63 [-1.78]	-5.58 [-2.36]	-2.20 [-0.92]
smb	-29.46 [-9.24]	-30.87 [-8.39]	-39.23 [-9.99]	-30.75 [-8.75]	-34.33 [-8.30]	-36.83 [-10.86]	-31.40 [-7.97]
hml	-14.44 [-3.86]	-17.55 [-4.40]	-19.17 [-4.72]	-17.17 [-4.35]	-20.64 [-5.05]	-17.40 [-4.11]	-10.61 [-2.62]
rmw	44.29 [8.44]	56.33 [8.97]	78.00 [13.71]	62.69 [13.01]	71.23 [13.71]	70.23 [12.60]	48.72 [7.43]
cma	23.16 [4.25]	33.83 [5.87]	31.77 [5.34]	17.13 [2.70]	29.53 [4.97]	30.83 [5.28]	17.79 [2.90]
umd	5.19 [2.67]	8.29 [4.10]	9.97 [4.58]	8.83 [4.45]	8.55 [3.99]	7.96 [3.48]	6.38 [2.92]
# months	428	432	428	432	428	428	428
$\bar{R}^2(\%)$	77	73	73	74	73	73	78

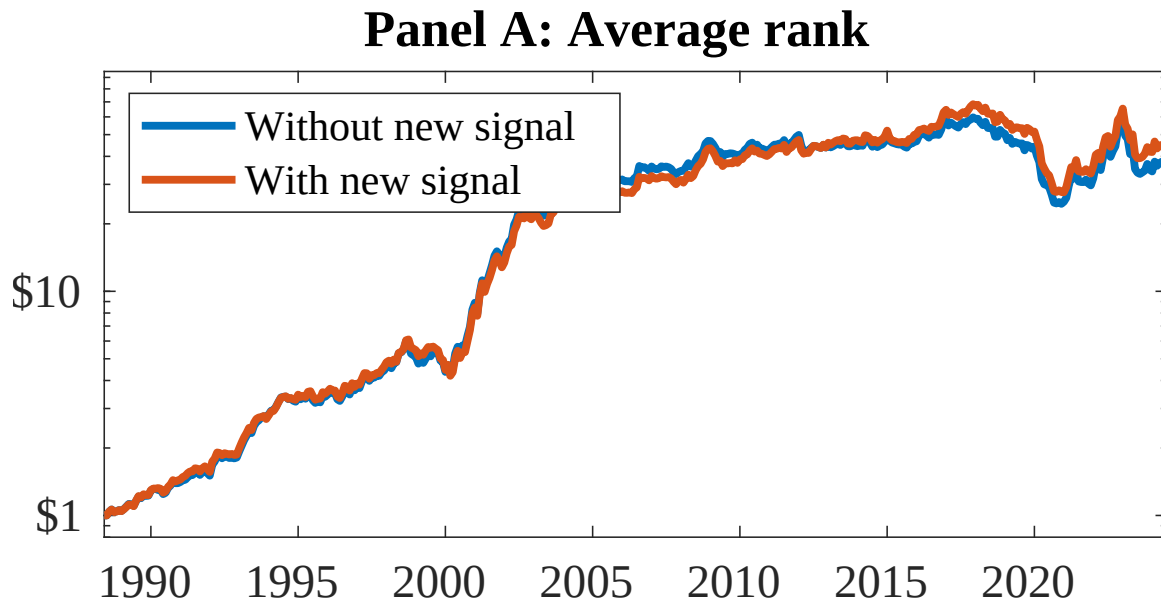


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ECG. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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